Coding and Solution Implementation

Trip Planner — Supporting Project Document

Frontend Request Lifecycle

The user completes the form, submits it, waits for the API response and views the returned content. The interface should prevent accidental duplicate requests where practical and show meaningful feedback.

Backend Request Lifecycle

FastAPI parses and validates input, constructs a prompt, calls Groq, processes the response and returns data in a predictable format.

Persistence Lifecycle

The save action submits the trip details and generated outputs. The backend maps fields to the database schema, writes a row and returns a success or failure response.

Implementation Quality

Validate input on the server, handle timeouts and external service errors, avoid exposing stack traces and keep secrets outside source code.

Maintenance

Keep dependency lists current, remove duplicate or obsolete source files and verify a production build after meaningful frontend changes.